

SAMPLE QUESTION PAPER

**CLASS IX
SCIENCE**

SECTION A

I. Write the correct answer (option a, b, c or d) in the box provided:

1. The following picture shows

- a) Dispersion of light
- b) Reflection of light
- c) Refraction of light
- d) Deviation of light



ANSWER

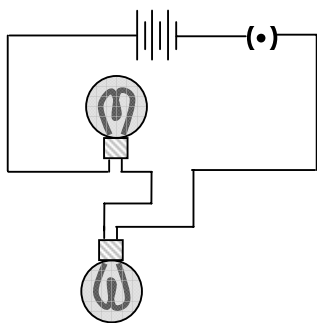
2. What is the type of lens used in a magnifying lens?

- a) Convex Lens
- b) Concave Lens
- c) Plano Concave lens
- d) Concavo Convex Lens

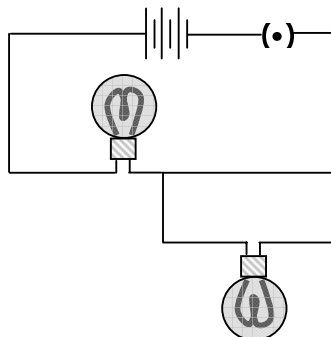


ANSWER

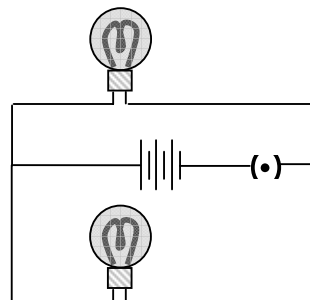
3. Which of the following circuit is a parallel circuit?



(a)



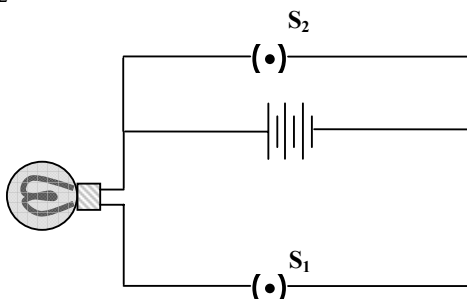
(b)



(c)

ANSWER

4. In the following electrical circuit, S_1 (a plug key) is switched on and the bulb glows. What will happen if S_2 is now switched on?



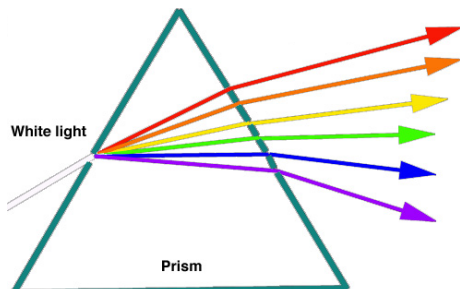
- a) The bulb will not glow
- b) The bulb will glow as before
- c) The bulb will glow dimly
- d) The bulb will glow more brightly

ANSWER

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II. Answer the following questions in brief:

01. The picture (below) shows a phenomenon of light. Name this phenomenon and explain why it occurs.



02. State any two properties of a bar magnet. Give one example where a permanent magnet and a temporary magnet are used.

03. What are fossil fuels? Give two disadvantages of using fossil fuels.

III. Numerical Problems:

01. Find the refractive index of the glass slab if the speed of light in the glass slab is 1.25×10^8 m/s. The speed of light in vacuum is 3×10^8 m/s

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02. In a given conductor, 10^{18} electrons flow through it in 1 second. Calculate the current flowing through the wire. Charge of one electron = 1.6×10^{-19} Coulomb

IV: Ray Diagram

Draw a ray diagram for a light ray refracting through a glass medium. Mark the following on the diagram.

- a) Refracted ray
- b) Emergent ray
- c) Angle of incidence
- d) Lateral displacement

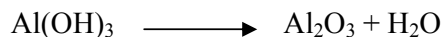
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SECTION B

1. What happens when magnesium burns in air?

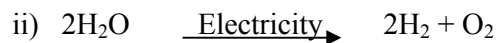
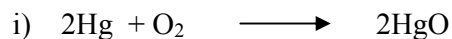
2. What is the relationship between an atom containing 11 protons, 11 electrons and 11 neutrons and another atom containing 11 protons, 11 electrons and 12 neutrons?

3. Balance the following chemical equation:



4. State two important properties in which diamond differs from graphite.

5. What type of reactions are represented by the following equations?



i. _____

ii. _____

6. Draw a labeled diagram of the apparatus used for the preparation of oxygen in the laboratory.

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SECTION C

1. A vaccine is usually given

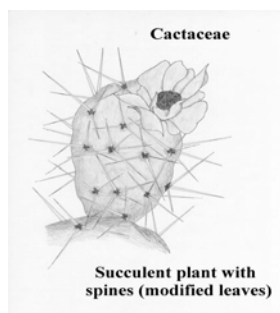
- (a) At any time
- (b) At the time of entry of disease causing germs
- (c) Before the symptoms of the disease appear
- (d) After the disease has appeared

ANSWER	
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2. Where will you find the plants shown in the figure-



A



B

- (a) A in desert; B in water
- (b) A in water; B on land
- (c) A on hills; B on plains
- (d) A in water; B in desert

ANSWER	
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3. **HIV** stands for-

- (a) Human immune virus
- (b) Human improper virus
- (c) Human immunodeficiency virus
- (d) Human immuno vector

ANSWER	
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4. Which of the following contains chlorophyll?

- (a) Yeast
- (b) Algae
- (c) Fungi
- (d) Mushroom

ANSWER	
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II. Answer the following questions

1. "Legume crops are included in crop rotation to build up the soil fertility".

(a) Explain the underlined word.

(b) How do legumes increase the soil fertility?

2. Give reasons-

(a) Earthworm is regarded as the farmer's friend.

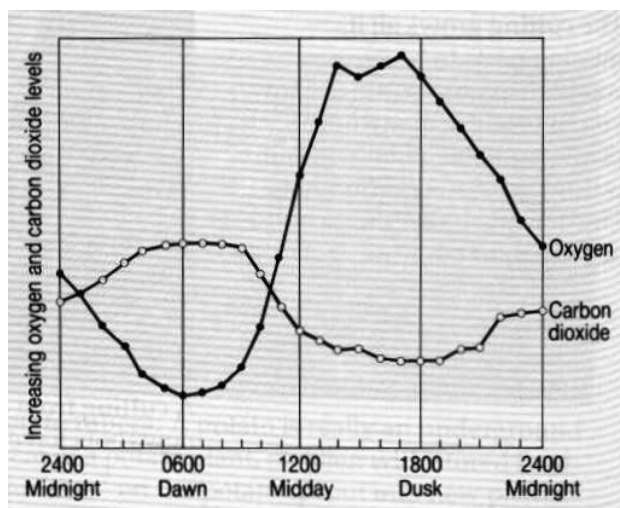
(b) Loamy soil is the best soil for growing plants.

3. Identify the agricultural practices depicted in the following pictures-



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4. A pond contains many water plants. The graph shows how the amounts of oxygen and carbon-di-oxide in the pond water change during the 24 hours from midnight onwards. Study the graph and answer the questions given below-



(a) When does the amount of oxygen start to increase?

(b) When does the amount of oxygen stop increasing?

(c) What happens to the amount of carbon-di-oxide in the pond, while the oxygen is increasing?

(d) Why does the amount of oxygen and carbon-di-oxide change in this way?
